



Parameter estimation of hydrologic models using a likelihood function for censored and binary observations



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ABSTRACT

Observations of a hydrologic system response are needed to accurately model system behaviour. Nevertheless, often very few monitoring stations are operated because collecting such reference data adequately and accurately is laborious and costly. It has been recently suggested to use observations not only from dedicated flow meters but also from simpler sensors, such as level or event detectors, which are available more frequently but only provide *censored* information. Binary observations can be considered as extreme censoring. It is still unclear, however, how to use censored observations most effectively to learn about model parameters. To this end, we suggest a formal likelihood function that incorporates censored observations, while accounting for model structure deficits and uncertainty in input data. Using this likelihood function, the parameter inference is performed within the Bayesian framework. We demonstrate the implementation of our methodology on a case study of an urban catchment, where we estimate the parameters of a hydrodynamic rainfall-runoff model from binary observations of combined sewer overflows. Our results show, first, that censored observations make it possible to learn about model parameters, with an average decrease of 45% in parameter standard deviation from prior to posterior. Second, the inference substantially improves model predictions, providing higher Nash-Sutcliffe efficiency. Third, the gain in information largely depends on the experimental design, i.e. sensor placement. Given the advent of Internet of Things, we foresee that the plethora of censored data promised to be available can be used for parameter estimation within a formal Bayesian framework.

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1. Introduction

Traditionally, mathematical models of river catchments and urban drainage systems are calibrated on uncensored observations of physical variables, such as discharge or water levels (Gupta et al., 1998; Madsen, 2003; Refsgaard, 1997; Westerberg et al., 2011). By uncensored observations we mean that the monitoring device used has an adequately large measurement range i.e. it can measure any physically possible value of the hydrologic variable. However, the installation and management of such monitoring devices for hydrologic systems is generally laborious and costly (Maheepala et al., 2001). Therefore, typically only a few locations are equipped with

such sensors. There are other types of sensors and observation equipment that collect data only within a certain range, and the value of any system response beyond an interval cannot be measured (Borup et al., 2015). Also, recent advances in sensor and data transfer technology development reveal a trend towards more cost-efficient sensors, which simply detect the occurrence of overflow events (Rasmussen et al., 2008) or detect only the exceedance after some critical water levels, thus providing binary information. Within this context, it has even been suggested to specifically develop binary monitoring devices based on robust and low-cost sensors such as temperature probes (Hofer et al., 2014; Montserrat et al., 2016, 2013), motion detectors (Siemers et al., 2011) and electrical switches (Rasmussen et al., 2008). Also, even though learning about the depth and velocity of flow from the footage of CCTV cameras during floods may be challenging, they easily capture the inundation and the duration of an event (Le Coz

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et al., 2016; Lo et al., 2015). In the context of Internet of Things, we are witnessing a growing trend where more real-time information related to hydrologic systems is available from hitherto unconventional sources (Eggimann et al., 2017; Kerkez et al., 2016).

Availability of any kind of relevant information related to the output of a modelled system should theoretically be able to reduce uncertainty in the parameters, allowing us to update our prior assumptions about their values (Riggelsen, 2006; Rinderknecht et al., 2014). As model predictions for hydrologic systems can be highly sensitive to various parameters (Gamerith et al., 2013; Song et al., 2015), the reduction in parametric uncertainty and identification of better parameter values has a direct bearing on the performance of models. It is therefore desired that in the future all types of data from different sensors, a) uncensored and b) censored, should help to calibrate hydrologic models within one consistent and general mathematical framework.

The underlying idea that many less accurate sensors may provide as much information about a complex hydrologic system as a few very accurate ones is very compelling. The first step to test this hypothesis can be to evaluate the value of binary observations in isolation, which can be seen as the most extreme case of censored data. While there has been research on modifying data assimilation techniques, like ensemble Kalman filtering, in order to handle censored data for updating the system state (Borup et al., 2015), the next logical step is to explore techniques that can infer model parameters from censored observations as well. However, so far only ad-hoc approaches to model calibration and parameter estimation using censored data have been suggested and it is currently not clear what the information content of such observations is and how we can use them most efficiently to learn about the system. Rasmussen et al. (2008) used the binary information measured during the occurrence of combined sewer overflows (CSO) to estimate the hydrologic reduction coefficient based on the mismatch between the predicted duration of the overflow and its observed duration. Similarly, measurements of overflow duration have been successfully used for calibration in other studies (Montserrat et al., 2016). Aronica et al. (2002) used a scalar performance measure and mapped it to the parameter space of the model with an informal likelihood function. The performance measure gives an aggregated indication of the performance of the inundation model. Thorndahl et al. (2008) applied similar mathematical framework for calibration of an urban drainage model and the quantification of parametric uncertainty using data on CSOs. In these two studies, those parameters that perform well on the chosen performance measure were identified as good and those which perform below a defined value were termed non-behavioural and not considered for model simulations. However, in the authors' opinion, this inference procedure is limited by the usage of an informal likelihood function (Stedinger et al., 2008). First, the criterion of acceptance/rejection of various parameter values is ad-hoc (Dotto et al., 2012). Second, having more observations does not reduce parametric uncertainty (Mantovan and Todini, 2006). And third, it assumes a perfect model structure and error-free input data, which is often not the case in rainfall-runoff studies (Del Giudice et al., 2015). To incorporate relevant error-generating processes, we need a formal likelihood function, which generally is lacking in the current treatment of censored observation. In this paper, we therefore suggest a likelihood function which makes it possible to estimate parameters from censored signals, while accounting for uncertainties in input variables, such as rainfall or land use, and model structure deficits. Based on the results from a case study, we demonstrate that censored data is surprisingly effective in reducing parametric uncertainty of rainfall-runoff models. Although uncensored observations are comparably more informative than censored data, our results are still quite promising, because they show the way

towards using all forms of available information in model calibration, which is often not done.

In the following sections, we first present the mathematical formulation of our likelihood function. We then demonstrate its application in inferring parameters and discuss why it requires a Bayesian framework (Section 2.1). The proof of concept is given through parameter estimation experiments for a real-life case study (described in Section 2.3). For this, we first infer parameters for a hydrologic model using binary data collected in our dedicated measurement campaign and then test the sensitivity of inference to experimental design and choice of priors (Section 2.4). In the same section, we also conduct tests on synthetic data from a simple linear model to facilitate reproducibility of this research. Finally, we present the results from simulation experiments (Section 3), provide a critical overview of this technique's limitations and interpret the results (Section 4). At the end, we draw our main conclusions (Section 5).

2. Method and material

In this section we lay out the mathematical formulation of our likelihood function. Once the likelihood function is defined, we discuss how Bayesian inference can be used to combine censored information and prior belief about the parameter values to update the probabilities of parameters. We then describe the case study and the simulation experiments carried out on it. The simulation experiments first show the performance of the inference procedure on binary observations. Finally we describe the sensitivity analysis to the threshold and prior.

2.1. Parameter estimation

2.1.1. Likelihood function

The explicit consideration of structural deficits and input errors in computer models, during the inference procedure of parameters, has been extensively discussed in literature (Higdon et al., 2004; Kennedy and O'Hagan, 2001; Todini, 2008). Reichert and Schuwirth (2012) propose to describe true system response Y_t at time t of an environmental system as the sum of a deterministic model m_t and a stochastic process B_t (see also Fig. 1):

$$Y_t = m_t(\mathbf{x}, \boldsymbol{\theta}^m) + B_t(\boldsymbol{\theta}^B). \quad (1)$$

We use the convention to represent random variables with capital letters and write vectors in bold. In Eq. (1), the stochastic process B represents the bias of the model due to structural limitations and input errors. The model inputs are denoted with $\mathbf{x}$ and parameters with $\boldsymbol{\theta} = \{\boldsymbol{\theta}^m, \boldsymbol{\theta}^B\}$, where the superscript m denotes the model parameters and B stochastic process parameters. It is reasonable to assume that observations $\mathbf{Y}^o = \{Y_1^o, Y_2^o, \dots\}$ of the system response contain observation errors, which are modelled by an independent additive error term (E_t):

$$Y_t^o = Y_t + E_t(\boldsymbol{\theta}^E) \quad (2)$$

For parameter inference, the probability (density) distribution of the observations given the parameters and input, $p_{Y_o}(\mathbf{y}_o|\boldsymbol{\theta}, \mathbf{x})$, the so called likelihood function, is required. It has been suggested to represent the bias B as a Gaussian process and consider independent normally and identically distributed observation errors E (Kennedy and O'Hagan, 2001). This makes it possible to write the likelihood function as a multivariate normal distribution with mean μ and covariance matrix Σ :

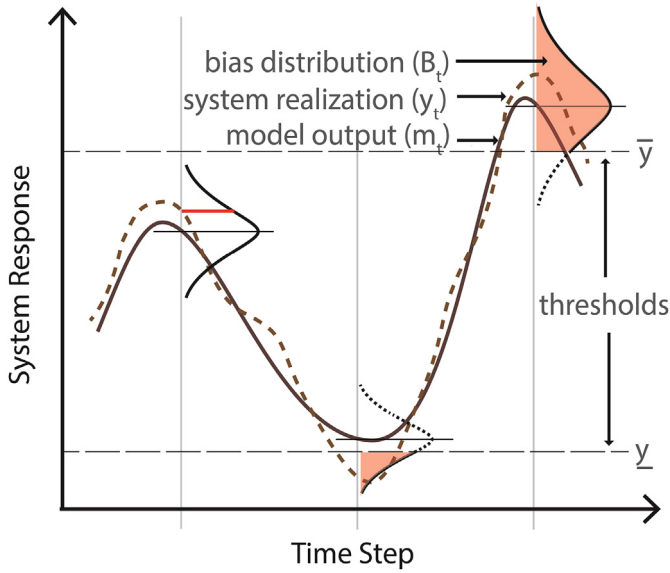


Fig. 1. Schematic representation of likelihood estimation for censored data. Integrals of bias distribution (shaded orange) need to be computed in the censored regions (below $\underline{y}$ and above $\bar{y}$), whereas the probability density is computed in between (red line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\mu = \mathbf{m}(\theta^{\mathbf{m}}, \mathbf{x}) \quad (3)$$

$$\Sigma = \Sigma_{\mathbf{B}} + (\theta_1^{\mathbf{E}})^2 \mathbf{I} \text{ with } \Sigma_{\mathbf{B}(i,j)} = (\theta_1^{\mathbf{B}})^2 \exp\left(-\frac{1}{2\theta_2^{\mathbf{B}}} |t_i - t_j|\right) \quad (4)$$

where $\mathbf{I}$ is the identity matrix, $\theta_1^{\mathbf{E}}$, $\theta_1^{\mathbf{B}}$ are standard deviations of the two error processes $\mathbf{E}$, $\mathbf{B}$ and $\theta_2^{\mathbf{B}}$ captures the degree of error correlation in time. However, the observation model in Eq. (2) is only adequate for uncensored observations. This prohibits using sensors providing censored observations, for example, measuring the system response only within lower and upper thresholds $[\underline{y}, \bar{y}]$. An observation of such a sensor can be described as (see also Fig. 1):

$$Z_t = \begin{cases} \text{AT} & \text{if } Y_t^o \geq \bar{y} \\ Y_t^o & \text{if } \underline{y} < Y_t^o < \bar{y} \\ \text{BT} & \text{if } Y_t^o \leq \underline{y} \end{cases} \quad (5)$$

i.e. the observation is either a measurement of the system response or it is “censored” and takes one of the special values: AT (above threshold) or BT (below threshold). Instead of discarding this information, the likelihood function defined by a multivariate normal distribution can be adapted to censored observations $\mathbf{Z}$ with an approach similar to what is used in survival analysis (e.g. Klein and Moeschberger, 2003). This is achieved by integrating the likelihood function over all possible values corresponding to either AT or BT (shaded areas in Fig. 1)

$$p_{\mathbf{Z}}(\mathbf{z}|\theta, \mathbf{x}) = \prod_{i=1}^n p_{Y^o}(y_i^o, \dots, y_n^o | \theta, \mathbf{x}) g(z_1, y_1^o) \dots g(z_n, y_n^o) dy_1^o \dots dy_n^o \quad (6)$$

where, n is the number of observations and

$$g(z, y) = \begin{cases} H(y - \bar{y}) & \text{if } z = \text{AT} \\ 1 - H(y - \underline{y}) & \text{if } z = \text{BT} \\ \delta(y - z) & \text{else} \end{cases} \quad (7)$$

is a helper function to ensure the correct limits of integration. The Heaviside function $H(x)$ equals to zero for negative x and one otherwise, δ denotes the Dirac delta function. Although the integrals in the likelihood function Eq. (6) are known to be analytically intractable for normal distributions, a very efficient numerical solution has been suggested (Genz, 1992). In this formulation, binary observations, such as those which we later use in our case study, are a special case of censored observations for which the upper and the lower threshold fall together, i.e. $\underline{y} = \bar{y}$.

This new mathematical description of a likelihood function generalizes its applicability to both uncensored and censored data. Akin to the case of uncensored observations, we expect the parameter inference to be more informative with increasing time series of censored observations, as the dynamics of the system are captured better. Also, binary observations that have multiple transitions from AT to BT and vice versa, would generally be more informative than those only having AT or BT as system response. This is expected because a relatively large section of the allowed parameter space can reproduce just AT or BT as a system response, which is not the case for specific sequence of AT and BT in the observations. After the likelihood function is explicitly defined, the parameters of the deterministic model and its bias stochastic process can be inferred within a Bayesian framework.

2.1.2. Bayesian inference

When the likelihood function is comprised of a deterministic model and an additive stochastic process to capture the remaining bias, a maximum likelihood estimation is not advisable. This is because the data can, in principle, be modelled equally well with both components, which leads to identifiability problems (Reichert and Schuwirth, 2012). This dilemma can be solved by introducing our prior beliefs on the parameters, for example that the bias has a high probability for a zero mean. This is possible via Bayesian inference, where our belief is formally expressed as the prior probability distribution of the parameters. In inference the posterior is proportional to the product of the prior probability and likelihood of observing the data, given the hypothesized model:

$$p_{\theta}(\theta|\mathbf{z}, \mathbf{x}) \propto p_{\mathbf{Z}}(\mathbf{z}|\theta, \mathbf{x}) p_{\theta}(\theta) \quad (8)$$

2.2. Performance assessment

To gauge the performance of the best parameter estimates prior to inference and posterior to inference, we compute their corresponding Nash-Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970). This performance index, which is a popular assessment tool for model performance in the hydrologic modelling community, captures the degree of mismatch between model prediction and uncensored observations. The absolute value of NSE after inference is limited by the performance of the underlying deterministic model. An inference technique can perform well even when the deterministic model is not capturing the system response with adequate accuracy. However, the improvement in NSE from prior parameters to posterior parameters does capture the performance of an inference technique. A performance index, based on similar philosophy as NSE, can be calculated for the binary system response, which tells us to what degree do the binary observations match with the binary predictions. *Percentage Matching Binary Observation* can be

defined as the number of time steps where binary predictions match binary observations divided by the total number of time steps. As we calibrate the model using binary observations, this index is expected to improve in the calibration phase. To quantify the reduction in the parametric uncertainty from the prior to the posteriors, we check the reduction in the spread on the marginal distribution of parameters. For informative data, posteriors are expected to be narrower than priors. Therefore, we analyse the ratio of the standard deviation of prior to posterior parameter marginal distributions. This contraction in the probability distribution of parameters gives an indication about how much has been learned about the parameter values using the new data of the system response. For a deterministic model, the posterior standard deviation of parameters should decrease as the time series of system response observation becomes longer.

2.3. Case study

We demonstrate the potential of censored observations by using them to calibrate a hydrodynamic rainfall-runoff model for a small urban subcatchment in the Wartegg city district of Lucerne, Switzerland.

2.3.1. Catchment description

The investigated catchment is a typical Swiss residential area with single to two-story housing of moderate density, including typical public infrastructure, such as a shopping centre, parking lots, and a school complex. The catchment area is drained by separate and combined sewers with a network length of 11.2 km, where an area of 30.2 ha is connected to the combined system. A CSO structure consisting of a side-flow weir and a retention tank with a storage capacity of about 100 m³ is located in the lower part of the catchment (see Figs. 2 and 3). In case of heavy rainfall, the CSO structure diverts excess discharge of combined sewage into Lake Lucerne. In this study, only the combined sewer system is

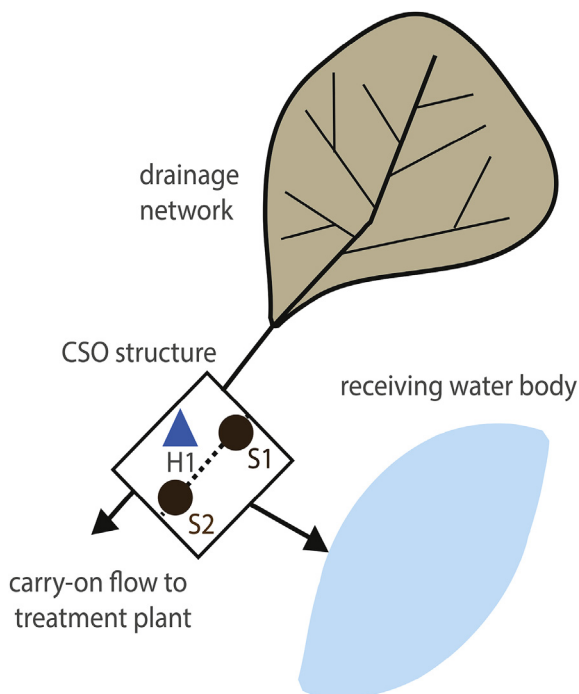


Fig. 2. Schematic outline of the experimental setup in the Wartegg subcatchment in the city of Lucerne, CH.

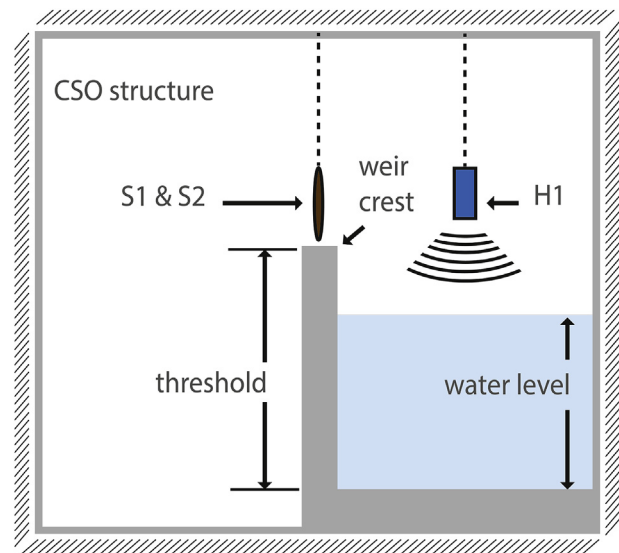


Fig. 3. Schematic of the experimental setup at the CSO structure in the Wartegg subcatchment: 1) H1 sensor: ultrasonic measurements (uncensored observations). 2) S1 and S2 TOMST moisture probe: binary observations. 3) Threshold is the weir crest. When the water level overtops the threshold, the S1 and S2 sensors capture the signal.

considered.

2.3.2. Uncensored data

We obtained the routine surveillance data from the utility, which operates an ultrasonic water level sensor positioned in the flow-through chamber of the overflow structure (H1, Fig. 2). The recorded water level H1, hereinafter referred to as uncensored data, is used as a benchmark for the binary data (S1 and S2, Fig. 2). In order to back up the operators' water level data H1, we additionally performed a targeted flow monitoring with two independent flow sensors about 200 m downstream of the overflow structure (carry-on flow) from July 2014–June 2015. Further details on this monitoring setup and instrumentation can be found in Peleg et al. (2017).

Rainfall data is obtained from a rain gauge located about 2 km west of the pilot catchment. The rain-gauge is operated by MeteoSwiss, the Swiss Federal Office of Meteorology and Climatology. It records rainfall heights at 10 min intervals with a precision of 0.1 mm, which we validated by hourly data provided through MeteoSwiss.

2.3.3. Binary data

At the overflow structure, two identical low-cost, multi-parameter sensors (TMS-3 data logger; TOMST, 2017) were installed, perpendicularly hanging above the weir crest to record temperature, moisture, and movement (S1 and S2 in Figs. 2 and 3).

The moisture signal (capacity-related measurement technique) clearly shows elevated levels when the sensor head is submerged, i.e. when the water level exceeds the threshold (Fig. 3). In the present study, the raw moisture signal was classified into either 'dry' (BT – no overflow) or 'submerged' (AT – overflow) conditions. The autarkic battery-powered sensors were permanently operated for two months from April until the end of May 2015, after their performance had been thoroughly tested and optimized at various locations over a period of several weeks. The recording interval was set to 1 min to sufficiently cover process dynamics.

2.3.4. Rainfall-runoff model

A hydrodynamic sewer model was developed based on

infrastructure data from the municipality's cadaster. The drainage system is modelled with EPA SWMM, version 5.1.009 (EPA, 2015) employing a conceptual hydrologic surface runoff model and routing the discharge in sewer pipes using the full set of Saint-Venant equations. The model physically consists of 158 sub-catchment units, 167 junction nodes, and 174 conduit links, which overall corresponds to a rather detailed catchment representation.

For parameter estimation, we first performed a sensitivity analysis to identify the most influential parameters: i) multiplicative factor for subcatchment width (θ_1^m), ii) maximum infiltration rate (θ_2^m), iii) and infiltration decay (θ_3^m), (both for HORTON infiltration method), which is in agreement with previous work e.g. (Liong et al., 1991). We then inferred these SWMM parameters together with those of the error model. The SWMM reporting step was chosen to be 5 min.

2.4. Simulation experiments

To be able to better interpret the performance and applicability of our inference technique, we performed simulation experiments for parameter estimation using the likelihood function described in Section 2.1 on 1) a simple linear model 2) and then on the case study described in Section 2.3. 3) We also conduct analysis to understand the dependence of inference on the value of threshold and the choice of priors.

2.4.1. Preliminary proof of concept using linear regression

As a preliminary test for the validity of this technique, we used a simple linear model with known “true” parameters. This model has been used before to compare the performance of various likelihood functions (Stedinger et al., 2008):

$$\mathbf{Q} = \mathbf{aP} + \mathbf{b} \quad (9)$$

where $\mathbf{Q}$ is the model output, $\mathbf{P}$ is the model input and $(\mathbf{a}, \mathbf{b})$ are the model parameters. Using the bias description from Eq. (1) to Eq. (4), we defined a likelihood function where the value of parameter vector $(\mathbf{a}, \mathbf{b}, \theta_1^E, \theta_1^B, \theta_2^B)$ is $(1, 1, 0.1, 0.3, 3)$. To generate synthetic uncensored observations, we used a sinusoidal $\mathbf{P}$, $\sin(0.1 \times t) + 1$, and sampled from the likelihood function. We then get binary observations by fixing a threshold of 2.5 for the uncensored observations. Data was generated for a thousand t values, from 1 to 1000. The prior belief for $(\mathbf{a}, \mathbf{b}, \theta_1^E, \theta_1^B, \theta_2^B)$ was defined using normal truncated distributions, having $(1.5, 1.5, 0.08, 0.2, 2)$ as means, $(1, 1, 0.1, 0.3, 1)$ as standard deviations and with the lower and upper bounds of $(0.25, 0.25, 0, 0, 1)$ and $(4, 4, 0.2, 2, 4)$ respectively. Parameter inference was done for the first five hundred binary observation values as the calibration event, and the inferred parameters were verified on the remaining time series.

2.4.2. Parameter estimation for the case study

We conduct parameter inference using real binary data obtained from our dedicated measurement campaign (see Section 2.3). These binary observations can contain false positives and negatives, and allow studying the quality of inference for real measurements as well as the robustness of the technique to noisy data. Based on the duration of our monitoring campaign for binary data, three rainfall events were used to assess the ability of binary likelihood function in inferring the model parameters, for which binary observations were available. The model was calibrated on one event, from 03/05/2015 12:00 to 05/05/2015 00:00, and then tested by predicting water levels for two different events, from 25/05/2015 03:00 to 27/05/2015 00:00.

For simplicity, while doing the inference, we only focus on $\mathbf{B}$ (Eq. (1)) and do not consider $\mathbf{E}$ (Eq. (2)). In our view, this is reasonable because i) model structure deficits and input errors are supposedly much more influential than random observation errors, i.e. detect/non-detects, ii) detectors of threshold exceedance are very simple devices which in turn are rather robust and, at least in our case, show very little randomness. While the optimization is done on posterior probability density of the parameters, we also compared the Nash-Sutcliffe efficiency (NSE) of the model that has been calibrated on the binary dataset to the NSE of prior parameters. As variables in hydrological systems, such as water levels in this case study, are often non-negative and errors are often positively correlated with the magnitude of the variable, e.g. large uncertainties with high flows/water levels, it is more realistic to first transform the system response with a function f , and then to apply Eq. (6) in the transformed space. This also requires transforming the thresholds $f(\underline{y})$, $f(\bar{y})$. For simplicity, we here suggest the Box–Cox transformation (Box and Cox, 1964)

$$f(y) = \frac{y^\lambda - 1}{\lambda} \quad (10)$$

which is commonly applied in many hydrologic case studies (Bates and Campbell, 2001; Del Giudice et al., 2013; Frey et al., 2011; Sikorska et al., 2012). Depending on the individual application, other transformations might be more realistic (Wang et al., 2012). The values for the Box-Cox transformation parameter was suitably chosen from the past experience in similar urban hydrologic studies (Del Giudice et al., 2013). The value of 0.35 produced satisfactory results for inference in the preliminary analysis. In this simulation, the model parameters that we calibrate are i) multiplicative constant for the width of the subcatchments (mult.width) and ii) the maximum infiltration rate (maxrate). We assume a prior distribution for our parameter space, capturing our best guess about the system. However, the prior distribution is kept wide so that we can learn from our measured data.

2.4.3. Sensitivity to threshold and prior

The effect of different priors was studied when inferring parameters. The standard deviation and mean of the prior was changed to gauge its impact on the inference procedure. The prior distributions should describe the knowledge of the analyst on model and error parameters, which also has to satisfy physical constraints, e.g. that negative pipe roughness or imperviousness values are impossible. Therefore, normal truncated distribution was used as the probability density function for priors, and five sets were chosen for the bounds, mean and the standard deviation for each simulation. We do not analyse the effect of using different types of probability distributions as priors e.g. lognormal, uniform etc. As in practice, different analysts often do not have the same level of expertise or available databases for different parameters, so we also alter the dimensionality of parameter space, by increasing the number of calibration parameters from 1 to 2 and then to 3, with different specifications of prior distributions (Table 1). The prior 1, 3 and 5 overestimate the peaks, whereas prior 2, 4 and 6 underestimate them. The uncertainty in the prior belief about parameter values is captured by the spread of the prior distribution. Wide as well as narrow priors were tested. The use of priors can help eliminate unrealistic parameter estimates and allows for the incorporation of accumulated knowledge about a system like a river catchment or a drainage network. As it is challenging to formulate the priors of the error model parameters, we have good experiences with defining them based on preliminary analysis of residuals of past observation periods. Besides, it is safe to assume that bias is unlikely to be higher than the standard deviation of the

Table 1

Information on priors used in the sensitivity analysis. All the priors of elements of θ are normal truncated. The vector in the table give the values for mean, standard deviation, lower limit and upper limit of the prior for each parameter in our inference. θ^m represents model parameters and θ^B the bias parameters in the transformed space (Eqs. (1) and (4)).

	θ_1^m (mult.width)	θ_2^m (maxrate)	θ_3^m (decay)	θ_1^B (sd.B_Q)	θ_2^B (corrlen)
Prior 1	[3,1,0.01,7]			[0.54,0.54,0,10 ⁸]	[0.05,0.05,0.01,0.1]
Prior 2	[0.1,0.5,0.01,7]			[0.54,0.54,0,10 ⁸]	[0.05,0.05,0.01,0.1]
Prior 3	[3,1,0.01,7]	[90,30,10,100]		[0.54,0.54,0,10 ⁸]	[0.05,0.05,0.01,0.1]
Prior 4	[0.1,0.5,0.01,7]	[37.5,30,10,100]		[0.54,0.54,0,10 ⁸]	[0.05,0.05,0.01,0.1]
Prior 5	[3,1,0.01,7]	[90,30,10,100]	[2,2,1,7]	[0.54,0.54,0,10 ⁸]	[0.05,0.05,0.01,0.1]
Prior 6	[0.1,0.5,0.01,7]	[37.5,30,10,100]	[2,2,1,7]	[0.54,0.54,0,10 ⁸]	[0.05,0.05,0.01,0.1]

system response itself.

Apart from doing a sensitivity analysis on priors, the general ability of the likelihood function to infer parameters for different thresholds was also tested using synthetic binary data. Simulations were conducted for the Lucerne catchment and the synthetic binary observations were constructed from the uncensored observations of water level H1 and five different thresholds (Fig. 3). Our prior knowledge on the model parameters was formulated as prior 1 (Table 1), then parameter inference was done using synthetic binaries corresponding to various thresholds. The effect of the threshold choice on parameter inference was studied using this preliminary analysis.

2.4.4. Implementation

We implemented the inference in the programming language R (CRAN, 2015) and used the *pmvnorm* function of the package *mvtnorm* (Genz et al., 2015) to compute integrals of the likelihood function. The mode of the posterior distribution in Eq. (8) can be estimated with an effort comparable to the use of an informal objective function. If the whole distribution is of interest, Markov Chain Monte Carlo (MCMC) methods are used to sample from the posterior (Vihola, 2012).

To speed up the inference, we constructed a fast surrogate of the SWMM model, a so-called “emulator” and used it instead of the

original SWMM model in the calibration phase (Carbajal et al., 2017). Although a single run of our SWMM simulator only takes about 20 s, this translates into an inference time of more than 55 h for an MCMC chain of 10,000 runs. In addition, the evaluation of the likelihood function requires numerical integration for each data point, which costs additional 10 s for each evaluation. To alleviate this cost, we followed the recipes in Carbajal et al. (2017) and built a data-driven emulator using the non-negative matrix factorization approach (more information in supplementary material). We produced 500 simulations with parameter values randomly drawn from the prior distributions and use the first 250 as training examples. The emulator prediction was evaluated in the remaining 250 simulations, achieving relative root mean square errors below 2%. The emulator demands 0.04 s per simulation, 500 times faster than the simulator, and reduces the simulation cost of the mentioned MCMC chain from about 30 h to 5 h, given the same overhead computational costs for both simulator and emulator to evaluate the integral in Eq. (6).

3. Results

3.1. Parameter estimation for the linear model

We found that the synthetic binary data was able to guide the

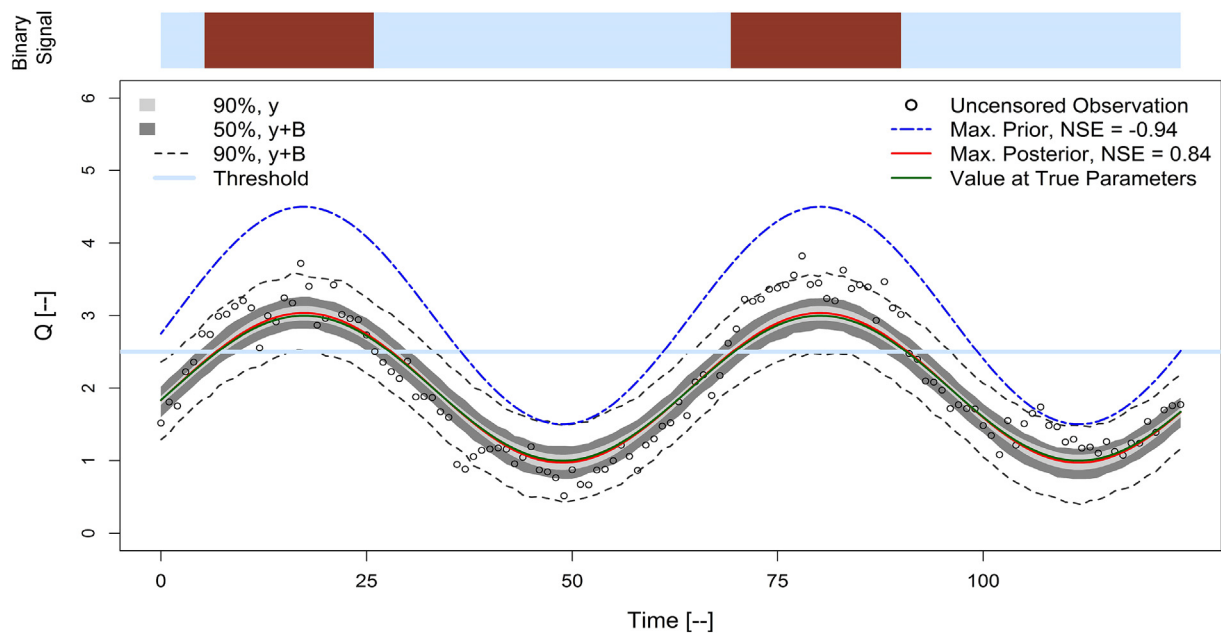


Fig. 4. The improvement in the model performance in the validation phase after inference using five hundred binary data points. The dashed blue line is the model prediction with the prior best estimate of parameters and the red line with the posterior best estimate. The y band (light grey area) represents the 90% interval generated after sampling from the posterior parameter distribution. The dark grey band and dashed black lines represent the 50% and 90% intervals respectively for $y + B$. Refer to Eq. (1). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

inference procedure towards the true parameter values. The maximum prior probability density parameter values that had a poor NSE (-0.94) were updated to parameter values that produce a significantly better NSE (0.84), after inference where five hundred data points were used (Fig. 4). The maximum posterior density parameter vector produced after inference is closer to its true values ($a = 1.03$, $b = 0.97$, $\theta_1^E = 0.075$, $\theta_1^B = 0.29$, $\theta_2^B = 2.03$). We found that the dependence in the joint density of parameters a and b was captured, showing a negative correlation (Fig. 5). We were also able to reproduce the decrease in parametric uncertainty with increasing time series of binary data used for inference (from hundred to thousand data points), with a decrease of around 20% in the standard deviation of posterior marginal distribution of the model parameters.

3.2. Parameter estimation using the case study

The main aim of this simulation experiment was to analyse the potential of binary data in directing us from poor prior estimates of parameters (fully uncalibrated model) towards better posterior estimates using real binary data (Fig. 6).

For this model, we see that the parametric uncertainty reflects itself much more pronouncedly in the peaks of the water levels, and does not influence the base levels much. The prior parameters grossly overestimate the water level peaks in the validation phase (Fig. 7). We see that the posterior obtained using uncensored data is narrower than the posterior using binary data, which is in turn narrower than the prior (Fig. 8a). The shrinkage in posterior captures the information gain in parameter values, and we also see that the joint distribution captures the dependence between various parameters (Fig. 8b). After calibration, the parameter estimates are taken away from regions which grossly overestimate the water level and higher probability is given to sections of parameter space which make model predictions that are in conformity with binary observations. We also see that the uncertainty bands stay wide as the inference is not provided with the uncensored data, therefore staying underconfident in defining the uncertainty intervals. Parametric uncertainty can be further reduced by utilizing newly obtained observations. The posterior distribution from one calibration event can be used as a prior during further inference on new events.

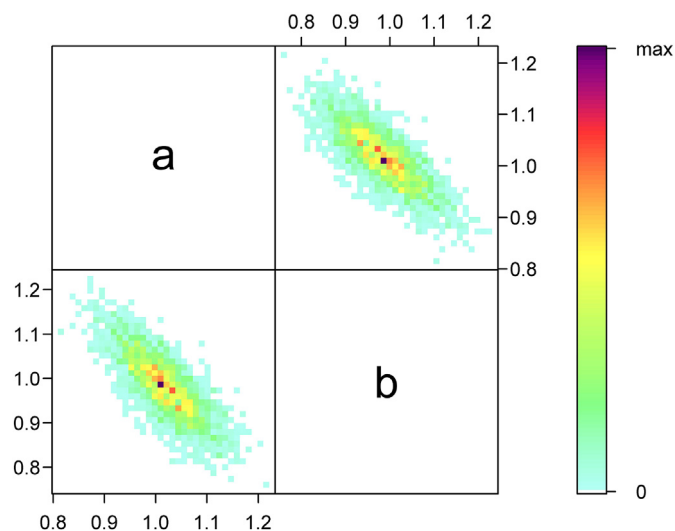


Fig. 5. Bivariate posterior distribution of model parameters a and b . The negative correlation that is not assumed by the prior gets captured in the posterior.

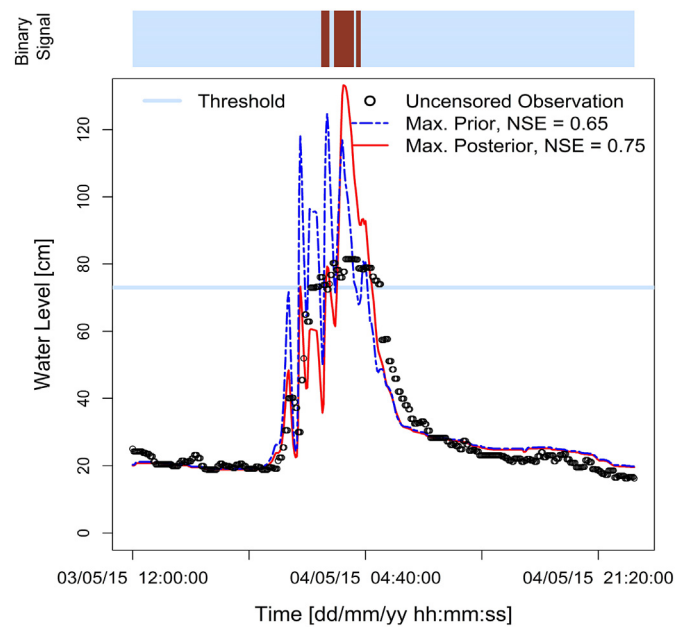


Fig. 6. The improvement in the model performance for the calibration phase after inference using binary data.

3.3. Sensitivity to threshold and prior

We find that for both calibration and validation periods, predictions based on the posterior outperform predictions based on prior knowledge. However, the quality of parameter inference is sensitive to the prior, i.e. the knowledge of or information available to the analyst. This is an expected result in Bayesian inference, as posterior probability distribution is, by definition, proportional to the product of prior and the likelihood. We find that the parameter posteriors are always narrower than their priors, thus greatly reducing the uncertainty in parameters (Fig. 9).

The inference leads to better parameter values, even when multiple parameters are inferred simultaneously (Table 2). This dependence on priors is expected to be pronounced for short calibration periods; theoretically, as longer and longer time series of data is used for inference, the impact of the likelihood function should outweigh the impact of priors in determining the posterior. Also, the posterior distribution generated gets narrower with increasing time series. However, as expected, inference using uncensored data for the same calibration period is much more informative than the binary data, especially in narrowing down the posteriors. Given the fact the binary information does not capture the detailed dynamics of a system response, the uncertainty intervals stay underconfident.

For varying threshold, the maximization of posterior distribution from the calibration event using synthetic binary data shows that the NSE of the posterior is higher for those thresholds which have a proportionate ratio of positives and negatives (Figs. 1 and 11). Thresholds which always (or never) indicate elevated water levels are less informative, or, in other words, virtually “blind” and thus do not contain much information to discriminate between likely and unlikely parameter values. The experiment was done using six different threshold values of water level; the observed maximum, minimum, and average water level for the event being 81 cm, 16 cm, and 32 cm respectively. This means that the threshold value below 16 cm will have only AT as binary observations and the threshold value above 81 cm will have only BT as binary observations. Given the fact that we have a fixed model structure, and we

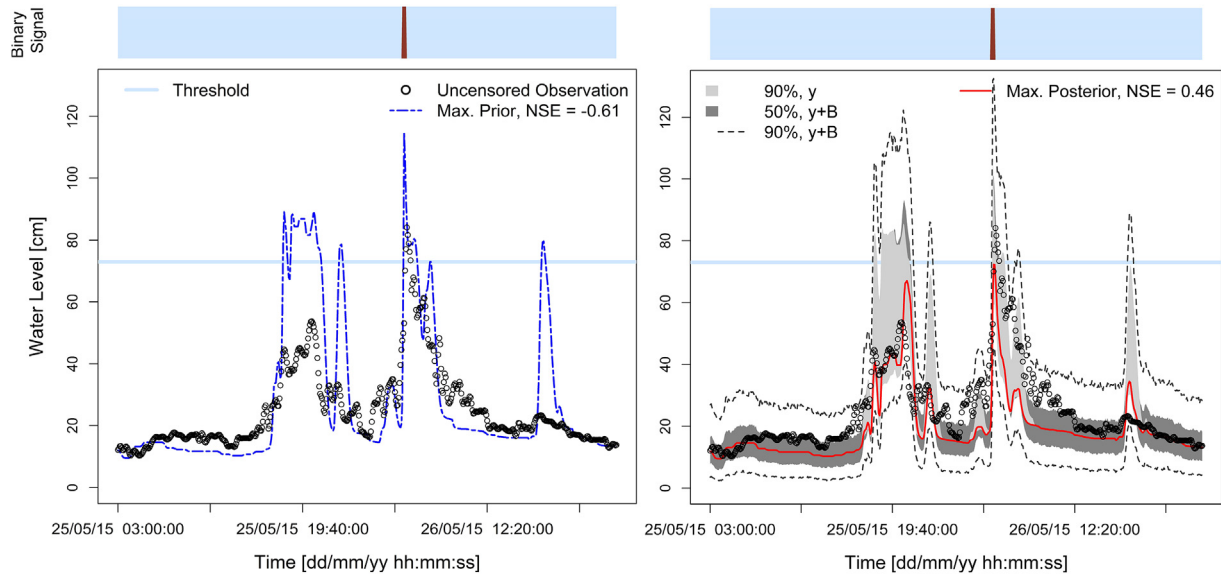


Fig. 7. The improvement in the model performance for the validation phase after inference using binary data.

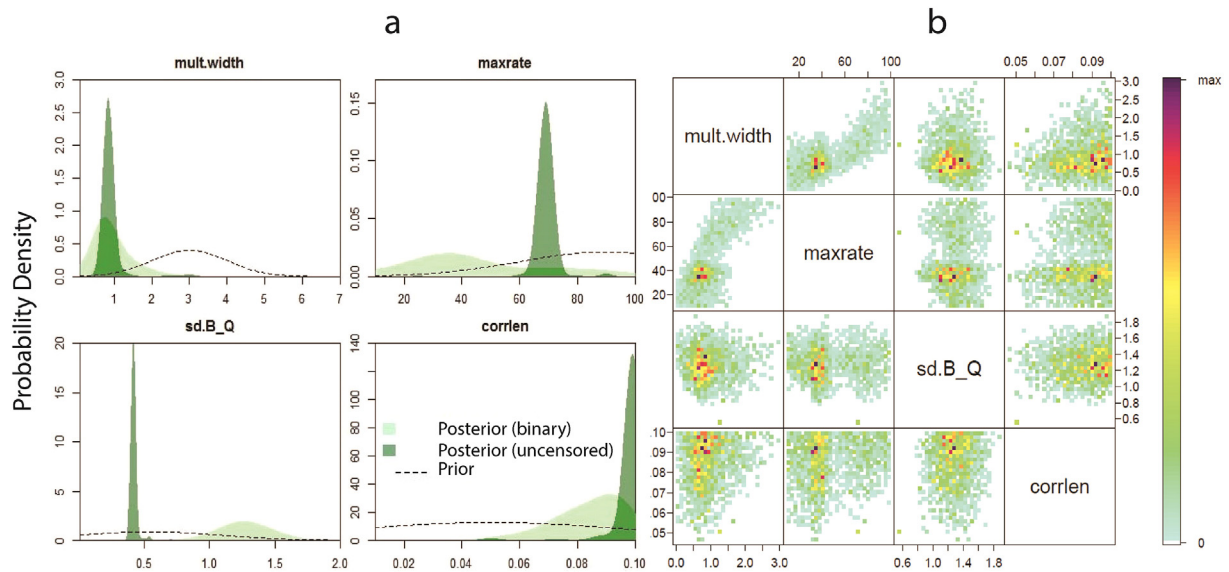


Fig. 8. Posterior distribution of model and bias parameters; (a) Marginal from binary and uncensored data, (b) Bivariate from binary data.

introduce a specific rainfall time series as input, the range of parameter combinations that can produce a particular sequence of binary signals is not wide. Therefore, while the information content of the signal containing only BT or AT is limited (Eq. (5)), the likelihood function is surprisingly useful to learn some information about the parameter values.

4. Discussion

The results of this research are in accordance with the theoretical framework that we employed. We expected noticeable improvements in model performance and our results confirm this. The main aim of this research was to formulate a formal Bayesian framework in which censored data can be used for parameter inference. It is difficult to properly associate probabilities to parameter values using ad-hoc methods. Also, it has not yet been

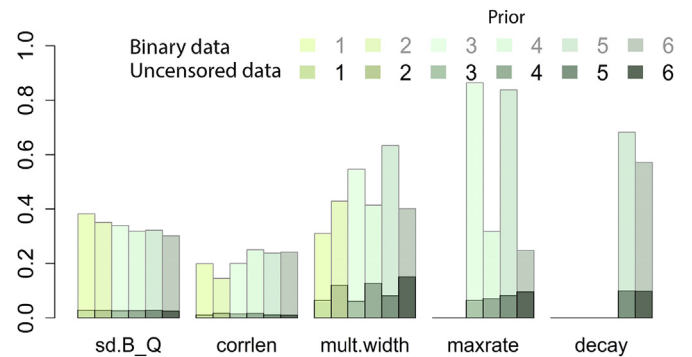


Fig. 9. Sensitivity to different priors and the gain of information from binary observations. The graph depicts the ratio of the posterior standard deviation to the prior standard deviation of parameters.

Table 2

Comparison of NSE for maximum prior and maximum posterior density parameter vectors (Table 1). Comparing the prior and posterior NSEs for the validation period shows the improvement in model performance after inference.

Simulation Type	Calibration Period		Validation Period	
	NSE (Max. Prior)	NSE (Max. Posterior)	NSE (Max. Prior)	NSE (Max. Posterior)
prior 1, one parameter	0.56	0.83	−0.61	0.32
prior 2, one parameter	0.47	0.82	−0.01	0.55
prior 3, two parameters	0.65	0.75	−0.61	0.46
prior 4, two parameters	0.65	0.64	0.02	0.04
prior 5, three parameters	0.65	0.78	−0.61	0.43
prior 6, three parameters	0.52	0.74	0.00	0.33

suggested how to combine uncensored and censored observations using such non-Bayesian inference procedures for hydrologic models. In the following sections, we examine the performance of a likelihood function that can integrate the inference procedure for uncensored and censored data in one consistent framework. i) We specifically discuss the information gained with respect to parameter values and the performance of the model with maximum posterior parameters. ii) We discuss the ability of censored data in the quantification of model uncertainty. iii) We interpret our findings vis-à-vis the dependence of inference on the choice of the threshold. We discuss the effect of priors on inference. iv) We also discuss the accuracy of binary observations.

Towards the end of the discussion, we talk about the big picture and make some recommendations on data collection and sensor placement so that data collected is better suited for calibration exercises.

4.1. Information gain from binary observations

As a simple validation of the likelihood function, Figs. 4 and 5 show that, for a linear model, inference using synthetic binary observations is able to reproduce the true parameter values reasonably well. For the real case study, where the assumptions of the likelihood function are only approximately valid, inference is still able to provide improved parameter estimates. Table 2 and Fig. 9 show that, despite the fact that the information content of censored data is inferior to uncensored observations, censored data in combination with a rigid model structure (SWMM in this study) and a well-defined input in the form of a rainfall time series allow us to learn about better parameter values. From all the different parameter combinations that the prior distribution allow, only a limited range of parameter values would produce the model output that agrees with the observed censored signal. Thus, the maximum posterior parameter values perform noticeably better than the maximum prior parameter values. In the presence of uncensored observations, our description of likelihood function evaluates the probability of observing a particular series of data, given the chosen SWMM and error model, for a specific combination of parameters. Whereas, when we have censored signals, the same description of likelihood evaluates the probabilities while taking into account a) the amount of exceedance/non-exceedance and b) the duration of that exceedance/non-exceedance. For example, the top of Fig. 6 shows an example of a binary signal and its duration. We see that the best estimate of parameter values after inference reproduce the observed binary signal more accurately than the uncalibrated model (Figs. 6 and 10). We also see that the calibration phase narrows down the parameter range after exploring the parameter space (Fig. 8). Parameters that cause longer durations of exceedance or shorter durations of exceedance are assigned smaller probabilities than those which correspond closely to the observed binary signal. Therefore, rainfall events that result in multiple instances of flow over the threshold will lead to better parameter estimation.

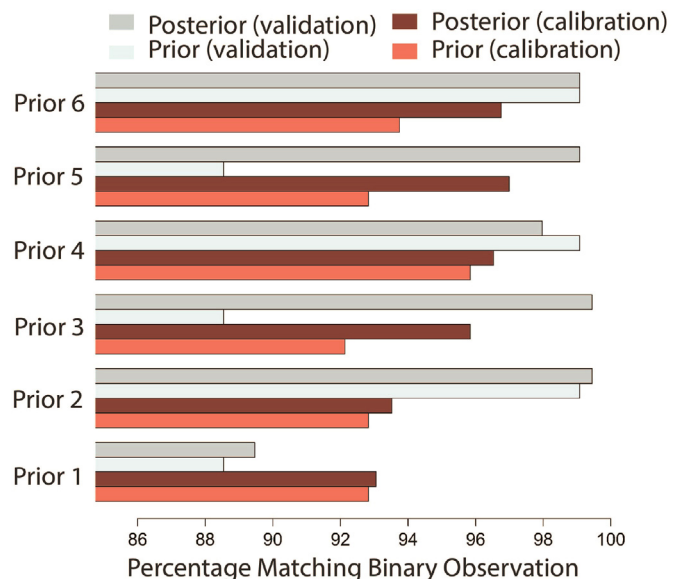


Fig. 10. Percentage Matching Binary Observation using the maximum probability density prior parameters and the maximum probability density posterior parameters. Inference in the calibration phase yields a noticeable increase in the model output matching with the binary observations.

4.2. Ability of censored data in the quantification of model uncertainty

Given a stochastic model, identification of the “true” parameters using a finite set of samples of the system response always comes with uncertainty. As the number of observations increases, the posterior probability distribution becomes narrower. This was also observed during the inference of the linear model parameters. The availability of censored data, in the absence of uncensored observations of the system response, still allows us to narrow down the parameter space and, interestingly, capture their dependence with each other, as it can be seen in Fig. 8. For hydrologic forecasting, the parametric uncertainty is usually found to be smaller than the model structure and input uncertainty when there is adequate reference data to calibrate the model (Del Giudice et al., 2015). Fig. 7 shows the uncertainty bands generated by sampling the parameters from the posterior distribution and running simulations using those parameter samples, labelled “90% y”, referring to the quantile and the deterministic model without bias respectively. As the binary observations are insufficient to assess the mismatch of model predictions and the actual system response, the standard deviation of the bias process is not reduced drastically through the inference procedure. This is expected, as binary data provides very little information about the actual system response, thus forcing the error model to prevent overconfident intervals. This results in uncertainty bands that are wide. In our view, it therefore seems best to

combine binary and uncensored data to reduce prediction uncertainty or utilize longer time series of data for inference.

4.3. Sensitivity to prior knowledge and threshold

The analysis also shows that the posteriors are sensitive to priors. Theoretically, this should be the case, especially if the event duration used for calibration is small. Bayesian probabilities represent the degree of belief and the posterior represents the combined effect of prior beliefs and the updating as a result of new information. Fig. 9 shows the reduction in parametric uncertainty after inference, for various priors. We see that different priors lead to different amounts of shrinkage in the density function of parameters, however, it does not hamper the inference procedure and the binary data is still able to guide us towards better posterior distribution. As far as deciding the prior distributions is concerned, we recommend not to exclude regions of parameter space that can potentially contain the sought parameter vector even though it is not highly likely, while choosing truncated distributions.

As far as the sensitivity to the choice of threshold is concerned, it is clear that signals with only BTs or ATs are the least informative. However, it is difficult to analytically formulate an exact relationship between the “value” of the censored signal in providing better parameter estimates and the ratio of BT and AT. This is because other factors, such as proximity of the actual system response to the threshold or the number of threshold crossovers, play a role. Our results show that the ability of the likelihood to learn about model parameters does depend on the threshold (Fig. 11). This can be explained, in part, by the high information content that crossing the threshold has. As the model has to be near the threshold when the observations flip from one side of the threshold to the other, the model learns more about the parameters which allow for such a model response. Thus, the ratio of positives and negatives and the number of cross overs in a binary signal will have a bearing on the information content of the data. In a real hydrologic network, different thresholds would correspond to installing an electrical switch or motion sensor, at different heights in the cross-section. Although in most real applications, binary sensors would detect overflow events at a certain weir with given height or a

predetermined critical levels that, for example, would lead to local flooding, we conclude from our first set of simulation experiments that the placement is most informative when a proportionate amount of BTs and ATs are detected. In real systems, the parallel installation of several sensors at different threshold heights may address this dependency. This seems most relevant regarding the experimental design for an entire network, where the information content from multiple threshold detectors can be maximized by installing them at different points and at different levels. In practical applications, the information content also depends on the state, i.e. level. For a utility manager, it is probably interesting to install all sensors at levels that correspond to 80% or 90% pipe capacity. Thus, continuously observing “no detects” in the entire network shows her/him that the network has sufficient hydraulic capacity.

4.4. Accuracy of binary observations

The performance of the binary sensors applied in this case study must be considered imperfect regarding sensor operation and accuracy. This holds true despite the fact that we installed two sensors of the same type in parallel (redundant application). The given binary observations are afflicted with noise, which is expected to influence the calibration performance. In this research, we did not include a specific error model of the binary observation instrument **E**, because a normal distribution is not necessarily a good error model for the noise. A dedicated error model incorporating an explicit false positive or a false negative rate for binary signals is relatively easy to formulate, however, the computational effort scales with 2^n , where n is the number of observations. Also, because the bias process **B** is generally capable of capturing remaining uncertainty, for example from noisy output observations, we do not see a practical need for greater detail in the error model. Ongoing tests with alternative low-cost sensor applications show that more accurate observations can be obtained compared to the accuracy of the data used in this study. This underlines the future potential of censored signals, in particular, when combined with low-power wireless data transfer techniques.

For measurement campaigns, we recommend obtaining a combination of uncensored and censored data. That can be a good trade-off between having only expensive accurate observations or only less informative censored signals. An optimal experimental design with binary sensors, however, has to consider their spatial location in the hydrologic network and the desired threshold levels. This goes beyond the scope of the article and requires further investigation.

5. Conclusion

We see that monitoring the dynamic response of hydrologic systems is often constrained by inadequately available manpower, technology and financial resources. Data from traditional flow and water level observation devices is hence limited. In the future, Internet-of-Things type communication platforms will also make it possible to operate low-power sensors that can transfer comparably little and often aggregated or censored data, such as the duration of an overflow or whether a threshold was exceeded or not. In this study, we suggest a suitable likelihood function to optimally use such sensor data in model calibration. This likelihood function can handle both uncensored and censored observations, without having to alter the underlying mathematical description of errors due to model structure and input. Based on theoretical considerations and the results from a real case study, where we predicted urban rainfall runoff, we draw our main conclusions:

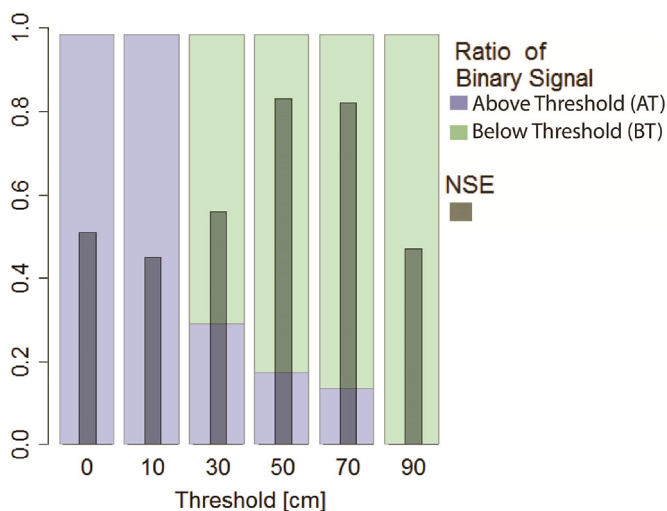


Fig. 11. The sensitivity of NSE to placing a binary sensor at different threshold levels in the cross-section. Results correspond to the parameter values at maximum posterior probability density in the calibration phase. Performance for thresholds that only contain AT or BT signals is poorer compared to other thresholds. Synthetic binary data corresponding to different thresholds were generated from the original uncensored observations.

- Given current developments in sensing, increasingly more sensors are becoming available, which often provide only censored observations. We suggest a likelihood function that is able to reliably learn about model parameter values from censored data.
- As our models and input data are always prone to deficiencies, we recommend choosing a likelihood function that can capture such deficiencies. One possibility is the usage of additive autocorrelated error models, which capture remaining uncertainties. Furthermore, using a Bayesian inference framework will help to avoid identifiability problems.
- Our results from a real-world case study in the city of Lucerne demonstrate that parameter estimation using censored data is beneficial, because the predictions are significantly better than the prediction with priors, with a change in NSE from -0.61 to 0.46 . We also find that the posterior parameter distributions are substantially narrower than the priors, thus reducing the model parametric uncertainty (standard deviation) on average by around 45%.
- We demonstrate that the inference procedure is sensitive to the choice of priors, which is expected when little or comparably less informative data is used.
- Using a likelihood function makes it possible to construct prediction intervals, which have a sound probabilistic interpretation. Although the uncertainty bands of the calibrated model are seemingly wide, they can be much narrower than those of the uncalibrated model. This is because a relatively short time series of binary signals should not be able to reduce the uncertainty in the model predictions significantly as only little information about the system is available. The likelihood takes that into account and keeps the bands wide.
- We find that, for having informative censored observations, a proper experimental design is crucial. This is because the location of the sensor, in the hydrologic network as well as within the cross-section (i.e. the resulting level of the threshold), affects the ability of the model to learn from such data. For model calibration, thresholds are advised to be chosen such that proportionate ratio of positive and negative signals can be observed.

To summarize, this research delineates a formal Bayesian framework so that we can utilize censored information, representing a variety of data types, for model parameter inference. We validate this concept using a synthetic and a real case study. The results show that the parameter estimation using censored data works quite well.

Author contributions

OW, AS, FB and JR wrote and structured the main text. OW conceptualized the simulation experiments, wrote the code, and carried out the analysis. OW and AS formulated the likelihood function. JPC built and trained the emulator, and also wrote the [supplementary section](#). FB planned and conducted the observations, and developed the rainfall-runoff model. JR and AS envisioned this project, whereas JR and FB actively supervised the research. All authors reviewed the paper.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.watres.2017.05.038>.

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